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Aspire! Afterschool Learning

McCourt School of Public Policy at Georgetown University Capstone Group: Aspire! Afterschool Learning Program Consulting Report

PREPARED FOR ASPIRE BY

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Executive Summary

Arlington County, Virginia is one of the most affluent counties in the country, yet there is a strong achievement gap between black and brown students from low resource families and their affluent, mainly white peers (Aspire, 2019a). Aspire Afterschool expands learning opportunities in an afterschool setting for underprivileged students, who are predominantly Hispanic and who speak another language at home.

To help Aspire gauge the effectiveness of their afterschool program, the Georgetown Team posed the following research question: What is the impact of attending Aspire on student reading instructional level outcomes as compared to similar students who do not receive this intervention? The Team hoped to find a comparison group based on the Aspire students' characteristics in order to create a counterfactual using a propensity score matching method. However, due to the lack a of individual level data needed to create a comparison group, the team was unable to answer the original research question and instead posed the following alternative questions: *How have students' reading levels and homework scores changed throughout the school year during which they participated in Aspire?* The Team was able to use descriptive statistics to come to the following conclusions:

- On average, students' reading scores increased throughout the year they were enrolled in the Aspire program;
- More than two-thirds of new Aspire students made substantial reading progress over the course of the year; however, returning Aspire students also showed widespread improvements;
- Results were mixed regarding whether new or returning Aspire students are more likely to have reading levels at or above their current grade levels;
- Although students' reading outcomes improved throughout the year, most students' homework scores remained unchanged between quarters one and four; and
- New Aspire students made more significant progress in improving their homework scores, and also had higher homework scores by the fourth quarter; whereas returning Aspire students, on average, saw comparatively lower homework scores and a slight decline in homework scores by the fourth quarter.

Going forward the Georgetown Team recommends that Aspire collect data on several new variables that will enable them to create an unbiased impact evaluation. These recommended variables, in addition to the data they currently collect, will be helpful in specifying future regressions and creating comparison groups. The recommended variables are:

- Gender
- Student race/ethnicity
- Standardized reading level
- School proportion of Free/Reduced Lunch students
- School racial composition
- Parental education
- Preschool attendance

In addition to more data, the Georgetown Team consulting report offers recommendations on how Aspire can adopt an implementation evaluation, develop social emotional learning activities, and establish a longitudinal study to assess the program's long-term academic impacts.

Aspire is committed to empowering Arlington's vulnerable students. This report places an increased emphasis on steps that Aspire may take in the future to continue to have a strong impact on Arlington County's less advantaged pupils. Assessing Aspire's statistical significance does face data difficulties, however with adequate changes this evaluation can be revisited.

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I. Introduction

Despite the wealth and resources in Arlington County, VA, the achievement gap between economically disadvantaged (mostly nonwhite) students and their more advantaged (mostly white) peers continues to grow (Aspire, 2019a). Aspire! Afterschool Learning (referred to as Aspire from this point forward), a 25-year-old non-profit organization, has worked to close the achievement gap by providing out-of-school programing for students at-risk of being left behind and expanding these students' learning opportunities. The only academic afterschool program for at-risk youth in Arlington County, the program is provided at one primary and three satellite sites in the Columbia Pike corridor, an economically disadvantaged area of the county.

Aspire's flagship program, Learning ROCKS!, delivers 15 hours of academic and socio-emotional support to 100 low-income 3rd to 5th grade economically disadvantaged students (EDS) afterschool each week, and in a six-week intensive summer camp, to bolster students' reading skills, and prevent summer learning loss among EDS students. Over the past five years, Learning ROCKS! students have achieved over 90 percent improvement in their reading instructional level (Aspire, 2019a).

Aspire supports the academic achievement and socio-emotional skills development of upper elementary students. It emphasizes the importance of literacy skills, as strong evidence shows that reading proficiency in upper elementary grade is significantly correlated with high school graduation, ultimately helping youth to fulfill their long-term potential.

Notably, Aspire has adopted an innovative, multilateral approach—after school, at school, at home—in its program intervention. Aspire instructors collaborate closely with teachers, school personnel and parents to support students' academic development and help establish healthy household relationships. Also, Aspire's unique Middle School Buddies Program uses existing middle school volunteers to serve as "peer mentors" and positive role model to younger students.

In the fall of 2019, a research team of students from Georgetown University's McCourt School of Public Policy began a quantitative research project for Aspire; to assess the program's impact on participating students' reading skills, compared to students from similar socioeconomic backgrounds that have not participated in the Aspire program, allowing Aspire to communicate these effects to stakeholders. The research team also looked for opportunities for Aspire program improvements to increase impacts on student learning and establish a more robust system for future evaluations.

The research project was driven initially by the major research question and two sub questions:

Research Question:

What is the impact of attending Aspire on student reading instructional level outcomes as compared to similar students who do not receive this intervention?

- *Sub-Research Question 1:* How should the evaluation project select similar students who are not exposed to the intervention as a comparison group?
- *Sub-Research Question 2:* What is the effect of Aspire program participation on reading scores?

In its literature review, the team examined afterschool programs in academic and non-academic settings to assess how Aspire's methods of a multilateral approach and group socialization can affect students' future success and narrow the achievement gap between advantaged and disadvantaged students in Virginia (Hanover, 2015). It also discussed the importance of various socioeconomic factors as well as socio-emotional learning in an educational context and developed information on what programs do in this field. The literature review also informed the analysis by providing insight on how other researchers have conducted similar studies.

Due to data limitations, the research question was ultimately revised to the following:

Revised Research Question:

How have students' reading levels and homework scores changed in the school year during which they participated in Aspire?

The change in question and its corresponding hypotheses are discussed in detail in Chapters III and IV below.

The paper is organized in six chapters. Chapter II is a review of the relevant literature. Chapter is the methodology and data description section, outlining the proposed question and potential analysis options. Chapter IV discusses our statistical analysis and its implications. Chapter V outlines several recommendations to strengthen Aspire's evaluation capabilities. And Chapter VI provides a concluding decision and a list of references and appendices that may prove useful to Aspire.

II. Literature Review

The literature review covers four areas directly related to Aspire's program and the program evaluation work our team conducted. We begin with an overview of research on the impacts of afterschool programs. This is followed by a review of literature on implementation of afterschool programs using a multilateral approach, which, similar to Aspire's program seeks to address several aspects that may affect student outcomes. We then discuss the importance of socioeconomic status, at the family, school, and neighborhood levels and their impact on academic outcomes on students. This provides us with theoretical justification in choosing potential covariates in our statistical model. We conclude this chapter with a review of the literature on socioemotional learning.

Afterschool Programs

Afterschool programs can positively impact students' academic, behavioral, and socioemotional outcomes. A wide array of the literature analyzed afterschool programs, both academic and non-academic in nature, and their effects on students. Research indicates that participation in academic-focused afterschool programs—such as Aspire's program—yields positive impacts on students' academic, behavioral, and motivational outcomes. The literature also defined the ideal parameters of afterschool programs, supported by empirical evidence, which can enhance the benefit of these programs for students.

Research on participation in afterschool programs has found positive impacts on students, especially students most at risk. For example, in a meta-analysis of existing quantitative and qualitative studies—including regression analyses and survey data—to define the prominent risks and benefits associated with afterschool programs, Cosden et al. (2004) found that afterschool programs specifically tailored for homework purposes, provided a myriad of benefits to students, including receiving homework assistance that may not be available to them at home, reducing stress so that students can be more relaxed at home, and reinforcing proper study habits and the importance of academic efforts (Cosden et al., 2004). Although the study also raised the possibility of afterschool programs causing students to forgo other activities promoting socioemotional development, such as reduced opportunities for parents to engage with their children on academic matters, a lack of coordination between the program staff and the child's teachers and defined curriculum, and an opportunity cost from children not having time to participate in other developmental experiences, the research on afterschool programs clearly pointed to more benefits than detriments for students.

In another study, findings indicated that gathering a smaller group of students in an afterschool atmosphere, as opposed to a full classroom size (i.e. a ratio of one instructor to many students) appeared to have positive impacts on students (Jones, 2007). Jones discerned substantial benefits of small group learning, particularly when such learning provided "opportunities for individuals to overcome barriers to learning caused by isolation" (Jones, 2007, p. 590). The study found that approaches taken to small group learning or teaching should reflect defined objectives for the students; such as paired, one-on-one discussion that promotes individual skills versus larger “buzz groups” that require more teamwork and thus can enhance social skills. This study is not focused specifically on elementary school students, which is the population served by Aspire, and further research is needed on the effects of various formats of small-group learning for low-income elementary school students.

As noted, a variety of literature exists on the positive effects of afterschool programs on students’ academic, and behavioral, socioemotional outcomes, including longitudinal studies that analyze the compounded success of students being in afterschool programs for multiple years. For example, afterschool programs that focus on reading attainment, such as Aspire, have been found to have a positive impact because they provide students with “the *opportunity* to learn to read” (Morris & Perney, 1990, p.134). The authors concluded that this opportunity to read aloud with correction is minimal in classrooms with one instructor and notably unequal across various socioeconomic groups, which is why afterschool programs can help amplify student reading success (Morris & Perney, 1990).

It appears that afterschool programs can have a compounding beneficial effect on students. In one study, using a quantitative analysis similar to the method our team used to assess the impact of Aspire's afterschool programs, researchers pretested students who would receive the treatment (the program) and those who would not (the control group). Researchers analyzed the data at one and two years of the afterschool programs, comparing the control and treatment groups, and found not only that the treatment students performed better in both years, but that the gap of increased success of the treatment group over the control group grew larger in nearly every category of the reading assessment (Morris & Perney, 1990). The program was similar in nature to Aspire, in that it focused on reading for second and third grade students. Despite a small sample of 30 students in Chicago, findings indicate that students who participated in two years of the afterschool reading program performed significantly better in the posttest assessment than those who did not participate.

Numerous academic benefits may result from collaborative learning or learning in which students work in groups among their peers. Specifically, Laal and Ghodsi (2012) discussed how this kind of learning setting has been found through multiple studies to foster critical thinking skills, require active participation from students, and help motivate students. Further, Laal found that, overall, collaborative learning promoted self-management among students and concluded that it “typically results in higher achievement and greater productivity, more caring, supportive, and committed relationships; and greater psychological health, social competence, and self-esteem” (Laal & Ghodsi, 2012, p.489).

Small group and individual tutoring methods may enhance students’ academic achievement, which proves beneficial for Aspire considering it incorporates both of these methods into its afterschool programs. Notably, existing literature has not established any significant differences in outcomes for students receiving small group academic assistance versus individual tutoring (Elbaum et al., 2000). The authors reviewed five programs created to assist first grade students struggling with reading (as determined by their standardized reading assessments) and found that students who received one-on-one tutoring from peers or older students saw greater academic outcomes than their untutored peers (Elbaum et al., 2000).

Studies have also attempted to discern whether or not afterschool programs can improve behavioral outcomes, given that at a minimum, they offer increased periods of supervision for students (Cosden et al., 2004). This concept is also highlighted by a U.S. Department of Justice study that found the greatest amount of juvenile crime typically occurs between 3:00 and 7:00 p.m., when students are home from school, but parents are typically not yet home from work (U.S. Department of Justice, 1999). Kremer et al. (2015) conducted a meta-analysis of 16 studies of afterschool program participation among at-risk, mostly middle school students, and found no significant differences on students' attendance or amount of problematic behavior (i.e. drug and alcohol use). However, they note that few authors of the studies contained in the meta-analysis attempted to reduce selection bias and they found other significant methodological shortcomings across the studies analyzed (Kremer et al., 2015). Considering Kremer et al. reviews much of the prominent existing literature on afterschool programs and behavioral outcomes, it is clear that more rigorous research is greatly needed in this area.

Educational scholars believe student motivation may positively contribute to both academic and socioemotional outcomes (Quirk & Schwanenflugel, 2004). The authors stated that motivation is crucial for reading attainment, as time spent reading is correlated to reading attainment. For students who struggle with reading, augmenting motivation is paramount to break the "predictable cycle of frustration, failure, and avoidance" they experience with reading. They find that steps to increase reading enjoyment, such as students choosing books they find personally interesting, matter little without first increasing motivation of struggling readers (Quirk & Schwanenflugel, 2004, p.7).

The study concluded that while most commonly used tactics within remedial reading programs improve students' motivation, programs could produce stronger motivation within students by incorporating elements such as goal setting and achievement among students. While the authors did not discuss specific examples of such goal setting, they concluded that the students should set the goals themselves to promote confidence and self-sufficiency and that the goals must be observable. Thus, such reading attainment goals could include completing a desired page number of pages read, learning to read and pronounce a certain number of new words, or learning a set number of new words of a higher reading level. The authors stated that the most effective method to address reading attainment is for program instructors to set aside time each day to individually converse with students about their self-set goals to determine "specific examples of observable increases in reading skill" (Quirk & Schwanenflugel, 2004, p.9). The authors found that this can increase students' motivation and assist them in focusing on their skills rather than comparing their reading skills to those of their peers.

In further support of the positive impacts of afterschool programs on student outcomes, a longitudinal study assessed the statistical impact of "high-quality" afterschool programs on 3,000 low-income and racially diverse students over a span of two years; of which 1,796 were elementary school students similar to Aspire's target demographic (Vandell et al., 2007). Notably, of these students, roughly 90 percent received free or reduced-price lunch and were students of color—again similar to Aspire's students. The authors measured the quality of the programs through a rating system that valued "evidence of supportive relationships" between staff members and students—as well as at a peer-to-peer level among students—and "evidence of rich and varied academic support, recreation, arts opportunities, and other enrichment activities" (Vandell et al., 2007, p.2). The authors rated in-person visits and conducted observations of the programs. Findings indicate that, among the elementary students who regularly attended the programs for two years, math attainment increased significantly, as did the students' "work habits and task persistence"—all in addition to their teachers' reporting significant gains in students' social skills and reductions in misconduct reports (Vandell et al., 2007, p.5). However, the study did not report on students' reading attainment, despite indicating that reading activities were part of the programs.

The reviewed impact evaluations of afterschool programs with a focus on math and/or reading showed academic improvements among students participating. Such programs, by nature, tended to address and increase student motivation to some degree. More research is needed to discern whether these programs positively impact students' attendance or other externalizing behaviors, although initial studies appear to show positive effects. A lack of literature comparing low-income student involvement in academic versus non-academic afterschool programs and these programs' comparative effects on student outcomes points to the need for further research in this area as well.

Multilateral Approach

Aspire engages in a multilateral approach to improving program participants' academic skills, including bilingual education, peer learning, and parental engagement. To fully assess the impact the program has on their participants, assessing each component of the program is important, understanding through the literature impacts found by other researchers and culling ideas for measurement of Aspire's unique approach. This section reviews literature on three aspects of a multilateral approach: 1) bilingual education, 2) peer learning, and 3) parental engagement.

Bilingual Education

Bilingual education is important to Aspire's curriculum because most of their students come from non-English speaking households and Aspire seeks to improve their English reading instructional level (Aspire, 2019a). A meta-analysis of 11 studies conducted in the United States indicated that students in a bilingual setting perform better on standardized tests in comparison to similar students who are only taught an English curriculum (Green, 1998). From the 11 studies analyzed, only five were randomized experiments. From those five, findings indicated a positive statistically significant effect on standardized test scores. The positive effect of bilingual education on scores for English tests was a .26 standard deviation increase, whereas the positive effect of bilingual education on scores for Spanish tests was a .92 standard deviation increase. In other words, a bilingual education setting is associated with an increase of standardized test scores – though, more so if the test given to the student is in their native language (Green, 1998). Strong evidence exists that bilingual education will help improve the reading scores of English as a Second Language learner, an instrumental component in Aspire's curriculum.

Peer Learning

Peer learning is a key component of Aspire's after school program. Research indicates that peer learning, where students engage in collaborative learning with their peers, is an important factor affecting student academic and social behaviors (Delay et al., 2016). Aspire is interested in measuring reading progress and assessing socio-emotional skills (Aspire, 2019b). While Aspire's program caters to third, fourth, and fifth graders, many students come back "to visit" once they age out of the program. Aspire is interested in adding "middle school buddies" to their curriculum and measuring program impact on these students (Aspire, 2019b). Research indicates that doing so may have positive impacts on the program participants. For example, Gray and Feldman (2004) found that having a mixed age group of students positively impacted social skills. The study found that older children's interactions with younger students encouraged their "social and moral education" by modelling social behavior and "articulating moral reasons [to the younger children] for behaving in one way versus another" (Gray & Feldman, 2004, p. 140). Moreover, the authors found that young children see adolescents as preferable role models over their adult instructors because their teen peers are closer to them in age and development stage. These positive effects support the idea that adding middle school students more formally to Aspire's program may increase positive outcomes of the program.

While Aspire's students can learn a lot from peers of different age/grade levels, they can also learn from students of different racial and ethnic backgrounds. Aspire's students come from a similar ethnic and socioeconomic background. There is evidence that a homogeneous classroom makeup is detrimental to these students' achievements. For example, one study found that reading and math scores dropped for both black and Hispanic students when in classrooms with a larger share of black or Hispanic students, respectively (Hoxby, 2000). It is important to consider this potential peer effect on Aspire's students' achievements as our team models program impacts, holding this variable constant to determine the effect the afterschool intervention has on students' progress.

Parental Engagement

A tenet of Aspire's methodology is to work directly with parents to ensure that the program targets other, non-school factors, that may impact a student's academic progress (Aspire, 2019a). To accomplish this goal, Aspire needs access to parents, but these parents face obstacles that hinder their ability to engage in their children's education and potentially their involvement in this afterschool program. Aspire's parents have several characteristics of "hard to reach parents," as most are low income and minority ethnic group (Hornby, 2011). Aspire's multilateral approach to improving reading skills depends heavily on communication between the program with both parents and teachers. While there is wide support for parental involvement in helping students academically, interesting research cautions practitioners about engaging parents in a way that is not useful. Edwards and Warin (1999) found that parents should not be used as teachers' assistants in helping their students complete their schoolwork (Edwards & Warin, 1999). The researchers strongly suggested against paternalistic training where parents are resistant to following authority. They also warn against requiring parents to teach their children a subject even though they may have no instruction in that subject. While Aspire seeks to encourage a healthy household relationship for their students (Aspire, 2019b) it is useful to keep parental limitations in mind when considering parental expectations and constructing parental training.

Measuring Success

Measuring the impact of a program is challenging. In this case, Aspire intends to measure the impact of their afterschool program on student's reading scores, and potentially other factors as well. There are various ways to measure "success" in different academic skills. The education researcher N.K. Lesaux (2012) argued that there is a difference in measuring “skills-based competencies” and “knowledge-based competencies,” and that students from low-income and non-English speaking backgrounds face great difficulty with the latter (Lesaux, 2012, p.73). Aspire is familiar with this subject in that they are targeting third to fifth graders because that is when they are “reading to learn” versus “learning to read” (Aspire, 2019b). Lesaux acknowledged that reading strategies must be changed in schools to allow students from vulnerable populations to exceed in future grades (Lesaux, 2012). Aspire’s focus on literacy skills and future academic achievement align with Lesaux’s findings.

Bilingual education, peer learning, and parental engagement are shown to have positive effects on students. However, more research to understand how factors of academic success are impacted by these different intervention methods is needed. Many studies vary in that there isn't a universal definition of academic success or what a bilingual education would consist of (Green, 1998). This makes it harder to apply a consistent form of teaching strategies across the board.

Socioeconomic Status (SES)

The United States is undergoing increasing income and wealth inequality, resulting in a significant and growing achievement gap between Black and Hispanic students and White students, and between economically disadvantaged students and their more advantaged counterparts (Coleman et al., 1966; Gonzalez, 2013; Lee & Burkam, 2002; Lynch &Oakland, 2014). Despite the wealth and resources in Arlington County, the achievement gap continues to grow (Aspire, 2019a). This section of the literature review delves into the key socioeconomic factors that contribute to the achievement gap. Previous research has recognized socioeconomic status as the best predictor of student’s educational attainment (Caldwell & Ginther, 1996). The conceptual definition of SES is multi-dimensional, including family SES, school SES and neighborhood SES. Recent studies also consider a fourth dimension—home resources (for example, home possessions related to learning, access to education services after school and in the summer) to be highly correlated with academic achievement (Sirin, 2005).

Family SES

Among the aspects of SES, family SES is the primary indicator of students’ achievement and explains the most variance (Coleman et al., 1966). Family SES is a tripartite variable including parental income, parental education and parental occupation. Each has been found to significantly correlate with educational attainment, alongside other associated factors intertwined with SES (Sirin, 2005).

Parental income: Parental income indicates the potential economic resources and social capital in the family that is available to the student. Duncan et al. (1998) found that among economically disadvantaged students, family income had the greatest influence on their academic achievement. The negative impact of early childhood poverty on child achievement was profound and cumulative, affecting the development of children's socio-emotional skills and cognitive ability. Family poverty in early childhood and the schooling years heavily influenced child achievement. Other studies have shown that low SES students tended to have low reading skills in kindergarten, leading to a higher incidence of poor academic performance in elementary, middle and high school (Duncan et al., 1998). Another study found that, compared with students who were never poor as children, those who experienced deep or consistent poverty for half their childhoods were nearly 90 percent less likely to graduate from high school (Smallwood, 2014).

These findings coincide with the present achievement gap both in Arlington County and Virginia as a whole (Aspire, 2019a) The negative impact of early childhood poverty on cognitive ability indicates the need for early intervention on economically disadvantaged students' literacy skills.

Parental education: Another stable indicator of student's future educational attainment and occupational success is parental education (Dubow et al., 2009). Taking into account the high correlation between income and education, parental education can serve as a key indicator of parental income (Hauser, 1994).

Parental occupation: Similar to parental education, parental occupation is a useful indicator for family income level (Sirin, 2005).

Family structure: A potential risk factor for family SES is family structure. Research shows that children raised by a single or divorced parent have lower educational attainment than children raised in two-parent homes (Cid & Stokes, 2013; Hampden-Thompson, 2013).

Primary language spoken at home: English proficiency is significantly correlated with students' literacy skills and better educational outcome (Glick et al., 2009). Research shows that at kindergarten level, students whose primary language spoken at home is not English had the lowest math and reading skill (Reardon & Galindo, 2009).

Free or Reduced-Price Lunch (FRPL): Several empirical studies have adopted student enrollment in FRPL as an SES indicator (Smallwood, 2014; Weber, 2012). FRPL is a widely used proxy measure of family poverty because it is a standardized requirement for income and an accessible way to acquire data. The Arlington Public Schools district policy is that students from a household whose income is at or below 130% of the federal poverty line are eligible for FRPL, which is useful in estimating the family income level (Arlington Public Schools, 2019). Further, FRPL longitudinal data are well-documented and easily acquired. However, despite its effectiveness, some researchers have concerns about its limitation, emphasizing that FRPL eligibility should not be considered the only measurement of SES for it could only capture part of the family characteristics (Snyder & Musu-Gillette, 2015).

For the Aspire project, FRPL eligibility can serve as an effective indicator of SES, since over 90 percent of Aspire students are enrolled in FRPL. Also, the unique lunch number of students helps trace longitudinal data when doing a follow-up survey on the Aspire program, such as the long-term impact on student's high school dropout rate. Notably, the proxy student data pool—Virginia Department of Education (VDOE) database—has flagged students as economically disadvantaged if they are eligible for FRPL or Temporary Assistance for Needy Families (also known as TANF). Although potential omitted variables bias exists because FRPL eligibility only captures family income and no other SES measures, the problem can be mitigated by introducing more proxy variables, such as race and English learner status, as SES indices.

School SES

The influence of school SES on the educational achievement of students continues to be viewed as a policy tool that addresses the achievement gap (Armor et al., 2018). A widely used approach to measure school SES is the percentage of students in the school eligible for FRPL (Leonard & Box, 2009; Sirin, 2005). The National Center for Education Statistics (2019) recognizes a school as high poverty when more than 75 percent of its students are eligible for FRPL (National Center for Education Statistics [NCES], 2019). The literature discusses both the negative impact of low school SES on educational attainment and provides ideas on ways to measure school SES.

Negative impact of low school SES on educational attainment: Lauen and Gaddis (2013) summarized two popular theoretical explanations for the negative impact on students' academic achievement caused by the high concentration of low SES students. One was the institutional factors theory, suggesting that lower SES public schools tend to have less qualified teachers, fewer school resources, and less engaged parents, leading ultimately to lower academic achievement. Another was the contagion mechanism theory, suggesting that SES effects have more impact at the classroom rather than school level because lower SES students have a lower academic expectation, poor study habits, and thus a more disruptive classroom culture (Lauen & Gaddis, 2013). An opposite theory, that low SES students in higher SES schools may have a more negative academic experience and increased psychosocial complications compared to those in lower SES schools, also exists (Crosnoe, 2009).

Another important risk factor correlated with academic achievement is school racial composition. Several studies have validated that substantial early achievement gaps exist between Black and Hispanic students and their White peers (Caldas & Bankston, 1997; Coleman et al., 1966; Reardon & Galindo, 2009). The racial/ethnic composition of a school, serving as a proxy variable for school SES, has a significant impact on individual academic performance, especially among economically disadvantaged students (Jaynes & Williams, 1989; Lee, 2007).

Neighborhood SES

Research has shown that even after controlling for the effects of family, school, and peers, the neighborhood SES is still another influencing factor on students' educational outcomes. (Martens et al., 2014). Children living in affordable housing generally have lower educational outcomes than their peers, while those living in affordable housing in wealthy areas have better outcomes than those living in poor neighborhoods. The concentration of poverty affects residents by limiting their access to social capital, including institutional resources, positive societal values, and potential neighborhood resources. The limitation of this argument is that it neglects the reluctance of some low SES families in choosing a higher-SES area to live and thus influencing the educational outcome of their children (Martens et al., 2014).

Aspire's programs operate in Arlington's Columbia Pike Corridor, an economically distressed area in a relatively affluent county. Most Aspire participants live in affordable housing (Aspire, 2019a). Evaluation of the program should account for the student housing as a baseline characteristic and take into account the neighborhood SES effect by comparing the Columbia Pike Corridor to other more advantaged neighborhoods in Arlington County.

Home Resources

Home literacy environment: A significant dimension of home resources in measuring socioeconomic status is the home literacy environment (HLE), which refers to the available literacy resources in the home (e.g. books, computer, study room) and cultural events (Buckingham et al., 2014). Research shows that lower SES households are more likely to have a negative influence on the development of early literacy skills due to lower-quality HLE (Cheadle, 2008). Thus, quality HLE is an important mediator for the relationship between low SES and student's early literacy skills.

Family educational investment: Promoting the literacy development of children is related not only to a quality HLE but also to a family's educational investment (e.g., reading-related parenting practices, attending regular parent-teacher conferences). Cheadle (2008) examined the significance of educational investment in improving children's math and reading outcomes and concluded that family educational investment was an important mediator of the socioeconomic and racial/ethnic gap. This finding raises the consideration that the impact of Aspire's program will not be restricted to students' reading achievement but will also include improving parental engagement, and hence their educational investment in their children's education.

Low Literacy Skills and Associated Risk Factors

Due to low literacy skills, low SES students may disproportionately experience low educational attainment and learning difficulties even before they enter formal schooling (Buckingham et al., 2014). However, the relationship between literacy skills and SES is not necessarily causal. Low SES can be a proxy variable for other variables (for example, preschool education, home literacy environment) which appear to directly influence children's cognitive development and educational attainment (Buckingham et al., 2014).

Aspire strives to close the achievement gap between economically disadvantaged students and their advantaged counterparts by improving their reading instructional outcome (Aspire, 2019a). Hence, this section limits the scope of discussion to factors highly associated with literacy skills and reviews several variables that either have an interaction with, or neutralize, the influence of low literacy skills caused by socioeconomic disadvantage.

Preschool education: The National Early Literacy Panel (2008) found that preschool may enhance the key emergent literacy skills of students by engaging them in diverse literacy activities (National Early Literacy Panel, 2008). Moreover, quality preschool education appears particularly beneficial to the most disadvantaged children, although low SES students who would benefit most from preschool education have the least chance of attending preschool (Brenchley, 2015).

Home literacy environment: A quality HLE is an important mediator for the relationship between low SES and student's early literacy skills. In the case of Aspire, early intervention could serve as an alternative for a weak HLE for low SES students, providing students with various age-appropriate books and group reading activities. Storch and Whitehurst (2001) found that children can benefit from group reading activities, which indicates that Aspire participants likely benefit from the group reading strategy and improve their early literacy skills, although this may be difficult to measure.

Primary language spoken at home: As noted, English proficiency is highly associated with students' literacy skills and educational attainment (Glick et al., 2009). Research shows that at kindergarten entry level, students whose primary language spoken at home is not English have the lowest math and reading skill (Reardon & Galindo, 2009). This factor is highly correlated with the Aspire program for more than 88% of Aspire students whose primary language spoken at home is not English. Moreover, over 50% of them have been enrolled in an English learner support plan (Aspire, 2019a).

Socio-Emotional Learning (SEL)

Socio-emotional learning (SEL), an important tool for youths' intra- and interpersonal development, promotes social and emotional youth development frameworks to reduce risk behaviors and amplify instruments for positive adjustments (Durlak et al., 2011). SEL is associated with students feeling more valued, having increased intrinsic motivation and, importantly for Aspire, developing a set of socio-emotional competencies that aid in achieving better academic performance and promoting healthy behavior (Greenberg et al., 2003). This section provides background on SEL that Aspire can use as a first step towards possibly integrating it into current programs and discusses key factors of SEL, including measurement indices, study outcomes, contextual factors, and the cost/benefit of SEL programs.

Indices

SEL encompasses a variety of evaluation frameworks, with researchers applying different indices as proxies to measure socio-emotional development. For example, a recent study by Kislyakov et al. (2016) attempts to establish a theoretical analysis of socio-emotional indices to evaluate the wellbeing of youth in an educational context. The study identifies three main indices for an educational environment: communication skills, social tolerance, and creativity (Kislyakov et al. 2016). Furthermore, a self-reported questionnaire administered to 700 university students finds a correlation between the formedness of personal qualities (the three indices outlined above) and subjective well-being.

Although the study outlines important aspects of socio-emotional wellbeing, it doesn't provide answers on how these should be weighed when designing intervention programs nor how to implement them. It is important to note that, while indices can be context-sensitive and differ depending on the program intervention, a clear body of evidence exists that associates SEL with improved academic, social, and other outcomes, regardless of the impact evaluation framework.

SEL Program Study Outcomes

SEL programs can differ on how they're delivered to students (dedicated class vs. supplemental program to core classes) and by whom (current teachers vs. external staff) among other factors. The literature on SEL programs indicates impacts on several important outcomes, including a student's academic achievement, success with transitions in school, and self-regulation and risk behaviors.

Academic Achievement: Numerous studies attempted to evaluate the outcome effects of different SEL program interventions on academic achievement in an educational context. For example, Durlak et al. (2011) conducted a meta-analysis of SEL programs, specifically in school-based universal interventions (program carried for the whole school student body). The analysis examined 213 schools and a total of 270,034 students from kindergarten through high school. Students received a variety of interventions, ranging from specifically developed SEL classes promoting socio-emotional competencies to multi-component SEL curriculum supplementing regular classes.

The study found that treatment groups (across all educational levels) exhibited an 11 percent gain in achievement and significant improvement in attitude, behavior, and social and emotional skills. Despite some inherent limitations to the analysis, namely 1) the focus on exclusively drawing research from universal school-based programs, 2) that a relatively small proportion of studies in the meta-analysis gathered information on academic achievement post-intervention, and 3) that a low proportion assess socio-emotional skills as a well-defined outcome; the study documented the clear benefits of SEL programs on outcomes similar to those that Aspire intends to improve through its afterschool programs.

Moreover, an increase in socio-emotional wellbeing is often associated with better academic achievement. Caprara et al. (2014) examined a program called "Promoting Prosocial and Emotional Skills to Counteract Externalizing Problems in Adolescence," a pilot school-based intervention promoting pro-social behavior curriculum for middle schoolers. Using a latent growth curve analysis, the intervention decreased aggressive conduct (both verbally and physically) and increased helping behavior, as well as enhancing academic achievement in the treatment group (Caprara et al., 2014).

The intervention included using prosocial sessions, a dedicated class featuring group discussions, and prosocial lessons, where teachers integrated concepts of the curriculum within their subjects during normal classes. The main components of the curriculum involved sensitization to pro-social values, development of emotional regulation skills, development of perspective-taking skills, and improvement of interpersonal communication skills. The study, however, used peer-rating measures to evaluate behaviors and employed a slightly older age group than Aspire's beneficiaries. Still, the curriculum and the study remain relevant to Aspire's multilateral intervention and interest to evaluate and quantify SEL effects, and may be something that Aspire wishes to examine further.

School Transition: Research indicates that SEL deficiency can lead to negative educational outcomes. For example, in a four-year longitudinal study of 8,568 nine-year old children, Smyth (2016) examined the factors affecting students transitioning from primary to secondary school in Ireland. The study also looked at immigrant students and the adverse effect of less culturally specific information about schools and curriculum. Findings highlighted specific socio-emotional factors hindering students from transitioning, including lack of emotional ties with teachers and failure to build a network of friends. A small percentage of students did not transition to secondary school at age 13 and thus excluded from the study. Failure to include this demographic in the analysis can overlook potential issues relating to their transition and outweigh other factors (Smyth, 2016). However, the assessment of social factors in transitioning to secondary school, and the separate analysis of immigrant students, may be useful to Aspire as they also serve immigrants in the same age group.

Self-Regulation and Risk Behavior: Research in the area of SEL outcomes showed that self-regulation and risk behavior are key outcomes of SEL, especially for youth. Aspire's beneficiaries can be considered at-risk youth, and programs that seek to improve their self-regulation and reduce their risk-taking behavior appear to have positive impacts on youth. For example, one study examined the role of self-regulation among children aged 6-13 on prosocial behavior, finding that although the majority of children embrace the norm of sharing resources, some fail to follow through. This gap correlates with self-regulation skills, namely attention and inhibition (Blake et al., 2015). Similar findings on a training program were found as effective so long as the treated individuals possessed high socio-emotional skills (mainly defined as empathy and consistency of effort). That study used a randomized control trial to evaluate the program impact of an at-risk youth training program in Brazil on various risk behaviors, including smoking, alcohol consumption and crime victimization (Calero & Rozo, 2016). Although the impact evaluation is not focused on an educational setting akin to Aspire's, the findings pointed to the long-term effects of SEL at an early age on future outcomes.

Contextual Factors

Afterschool Programs: The setting of SEL interventions affects resulting outcomes, with some studies focusing on an alternative setting to the normal school hours. Not only are these programs getting more attention as a way to address student gaps, but it is also important to put SEL in the context of afterschool and/or out-of-school programs to draw similarities to Aspire's intervention. A dissertation by Blattner (2018) looked at out-of-school programs as an answer to the growing achievement gap and analyzes their socio-emotional climate. Using social competence and resilience (measured through a specialized version of the Devereux Student Strengths Assessment) as the outcome variable, the study found four "GROW" dimensions of socio-emotional climate: Growth-promoting instructions, Resolve and focus, Organization, and Warmth.

Employing a hierarchical linear regression, the socio-emotional climate explains a statistically significant relationship to students' social competence and resilience, more so than demographics. The study used secondary RCT data of 2,500 rising fourth grade students in 37 summer learning programs. As such, the author relied on RCT's data collection, which mostly included binary (yes/no) answers and failed to explore other qualitative variables (Blattner, 2018). Nonetheless, given the focus on out-of-school programs to develop socio-emotional skills, this is relevant to Aspire's desire to reduce the achievement gap.

Mentoring: Socio-emotional skills are shown to be highly associated with strong connections to mentors, providing another building block to successful SEL interventions. A research study by Hurd et al. (2013) examined the relationship between involved-vigilant parenting (mentoring) and socio-emotional wellbeing among black early adolescents. Subjects with a strong connection to their mentor displayed higher socio-emotional wellbeing and skills than those with weaker connection or no mentor. The study sampled 259 early adolescents from three different middle schools and used k-means cluster analysis to identify the strength of the mentoring relationship. The mentoring relationship was determined by relationship length, involvement, closeness, and frequency of contact (Hurd et al., 2013). The study did not identify directions of causality between parenting, mentoring and socio-emotional wellbeing as it was a cross-sectional analysis. Aspire's intervention is based on the active involvement of staff during and after school and at home. This could create a similar mentoring relationship to the one discussed here.

Housing: Research shows that housing quality impacts a child's socio-emotional wellbeing. Employing the Children's Behavior Questionnaire (CBQ) to measure the socio-emotional health of 95 Canadian children, Gifford and Lacombe (2006) analyzed the relationship between physical housing and neighborhood quality and the socio-emotional health of children aged 9-12, while also assessing the housing and neighborhood quality using up to 309 aspects through parent interviews and walk-throughs.

Findings indicated a significant relationship between these variables, where worse physical housing and neighborhood conditions corresponded to lower levels of socio-emotional health. This relationship remained significant even after controlling for income, gender, time lived in residence, and parental education and mental health (Gifford & Lacombe, 2006). Study limitations included the failure to account for adverse parental behaviors, such as alcohol abuse, among other possibly confounding variables. Still, this example points to the importance of housing and neighborhood effects on socio-emotional wellbeing, relevant as Aspire actively promotes healthy households for their beneficiaries and their parents, who predominantly live in affordable housing in an economically distressed area (Aspire, 2019a).

Costs and Benefits

Understanding the costs and benefits of SEL programs is vital in assessing the feasibility of interventions. Belfield et al. (2015) evaluated four SEL programs and conducted a benefit-cost analysis (BCA) to determine their economic value. Results pointed to these programs having benefits that significantly outweighed the costs. The authors discussed several important distinctions between programs in calculating costs, mainly whether interventions are supplemental to normal classes or exist as a separate class. SEL classes are stand-alone, separate classes where students are specifically taught a socio-emotional focused curriculum. They are also associated with higher costs compared to supplemental classes (Gifford & Lacombe, 2006).

Given the varying nature of SEL program interventions, the direct costs and benefits are often hard to correctly quantify or measure. Calculations of benefits can be underestimated for disadvantaged students, who are likely to realize more benefits. Furthermore, some benefits might not have been priced due to the expansive and long-term effects of such programs. As Aspire plans to engage in evaluating SEL and possibly expanding their programs, the economics of such interventions should be taken into account.

Conclusion

The literature indicates strong support for the impact of afterschool programs and the use of a multilateral approach on improving student academic and behavioral outcomes, and the benefits of incorporating SEL. Through the literature, our team identified key variables useful for evaluating and measuring the impact of an afterschool program, including socioeconomic status and socio-emotional factors. The research reviewed shows that afterschool programs motivate students by providing learning opportunities outside of the classroom. Aspire cultivates an environment that encourages reading but there are likely numerous positive spillover effects beyond reading improvement that program participants experience. These should also be considered by Aspire for future assessment and evaluation as part of their program impact.

III. Methodology and Data Description

This chapter provides a description of the methods undertaken by the Georgetown team as we sought to address Aspire's research questions. We begin with a discussion of the intended quasi-experimental methods we planned to use to conduct the research; first propensity score matching, and then ordinary least square modeling. Due to data restrictions however, the team was unable to enact either approach and instead conducted descriptive analyses, described below in the section entitled Description of the Data.

Methodology

Propensity score matching method

Our team initially sought to evaluate the impact of Aspire on their students using propensity score matching (PSM), an effective approach to reduce selection bias and estimate unbiased treatment effects in a non-randomized study. However, PSM requires a series of baseline-observed characteristics at the individual level for both the treatment group and the comparison group. Due to a lack of individual-level data for the proxy students in Arlington Public Schools (APS), the team could not conduct its initially proposed quasi-experimental quantitative analysis. However, we have described the details of PSM in the *Alternative and Recommendations* chapter below, and listed recommended variables in *Appendix I* to help Aspire conduct such analysis in the future. Once Aspire obtains the permission to access the APS database and can acquire individual-level data from a wide range of student pools, it could conduct a quasi-experimental analysis using PSM and derive strong quantitative evidence to validate its program's effectiveness.

Ordinary Least Square regression model

Lack of individual-level data to create a comparison group, the Georgetown team attempted to employ an alternative analysis approach—Ordinary Least Square (OLS). This method also required data that did not exist in data set, including the independent variable data and important control variables necessary to run this this type of regression model. However, we provide Aspire with information about this approach when in case it proves useful to Aspire for future research. In our OLS regression model, our intended dependent variable would be the reading level reached by the students at the end of the year (these data are in the data set). Our independent variable ideally would be the dosage amount (measured as the number of days in attendance at Aspire) of the Aspire program. But that information is lacking in the data set. The key control variable, in this case, would be the initial reading level of the students. By controlling for the initial reading level and demographic characteristics that may affect the academic outcome, the model could capture the isolated impact of participating in the Aspire program on student's reading scores (Wilson, 2016). If used, the OLS regression equation would have been:

$$\gamma_i^{endread} = \beta_0 + \beta_1 Z_i^{startread} + \beta_2 attendance_i + \beta_N \chi_i^N + \varepsilon_i$$

Where:
i = individual
$\gamma_i^{endread}$= the student's reading level after program intervention
$Z_i^{startread}$ = the student's reading level before program intervention (baseline reading level)
$attendance_i$ = a prospective continuous variable measuring the attendance days of each individual per year
χ_i^N = a vector of control variables assumed to affect the outcome of γ, including grade, race, special education status, English learner status, duration of participation, site effect, school effect.
ε_i = a random error term

Because the Georgetown team did not have demographic information for all the students in the sample, nor the number of days that each student attended the Aspire program, we could not control for the variables necessary to yield a statistically significant result. Running a regression with the variables would have produced a heavily biased result without statistical significance that could not reasonably have been interpreted with confidence. Specifically, we could not use OLS regression to assess the impact of the Aspire program on student outcomes for the following reasons:

Selection bias: Due to the absence of a comparison group, there was not a sufficient counterfactual to assess the effectiveness of the intervention. The subgroup comparison between the Aspire students is inevitably affected by selection bias. In particular, the key independent variable—attendance days, is closely related to subjective characteristics. For example, this regression reveals that low-frequency participants (less than 90 days) outperform high-frequency participants (over 90 days) in terms of the degree of improvement. However, it is highly possible that high-frequency participants may be students who lag in their respective grade levels and require more study time, while low-frequency participants are those who fall behind grade level to a lesser degree. Hence, the ultimate improvement in reading level can be influenced by the preexisting subjective difference between the two groups and the OLS regression model with the included variables cannot parse out these differences.

Low sample size: A low sample size increases the difficulty of obtaining a statistically significant outcome; the chance of a result being due to coincidence is higher than if the model is run on a large sample. More cases in the data set will improve the model's ability to detect significant differences between students. A rule of thumb for sample size is 100 valid samples for both treatment group and comparison group.

Omitted variable bias: The challenge in a non-experimental study is to control for all extraneous variation that may influence the dependent variable—the academic reading outcome of participants. Aspire has comprehensive data on demographics and student reading scores in different stages, but more data on program implementation is needed to account for important factors that may be impacting how the program is conducted at each site. Differences across sites could also confound the estimation of the treatment effect. We further discuss the importance of collecting process variables in the *Alternatives and Recommendations* chapter below.

Data Description

Aspire Sample Data

The main source of individual-level data used in our research comes from the 2016-2017, 2017-2018, and 2018-2019 Aspire Results Spreadsheets. Our total sample size is 82 students. To arrive at this number, in order to ensure greater accuracy in our descriptive analysis, we removed any student record containing missing data, such as missing reading or homework scores for any one of the four school quarters.

Next, to eliminate duplication, we removed data for returning students, except for their latest record (i.e., if a student attended Aspire starting in 2017, only his or her record for the 2018-2019 school year was included). These actions ensured a sufficient mix of new and returning Aspire students for our study. There are 42 new students and 40 returning students in the sample—a roughly proportional combination. While unable to support a longitudinal study, our method of data selection adds somewhat of a longitudinal element to the analysis, as we separated these returning students and compared their achievements to those of new students. Should Aspire wish to collect the necessary information to conduct a complete longitudinal study in the future, the Georgetown team has included guidance on how this may be accomplished in the *Alternatives and Recommendations* chapter of this report.

The team had planned to include demographic information, such as race and free-or-reduced-price lunch variables in its analysis, but, we did not have timely or complete data for these variables for all students in the sample. Thus, our analysis does not include these variables. However, given that the majority of Aspire students are low-income, minority students, we felt our descriptive analysis would still provide an adequate view of how these students may be falling through the gaps at Arlington County Public Schools and how their academic attainment may change throughout the year with Aspire.

Originally, we had planned to assess students' reading and homework attainment, not merely as a whole, but also by school and/or learning site. However, our team decided that it was important to use the full 2016-2019 data set, which included sites phased out during the data collection for this time period. Aspire determined that the focus should be on the academic attainment of Aspire students generally, rather than separately by each site, with their much smaller sample sizes and less useful summary statistics.

New Variables and Descriptive Analysis

The team introduced several new variables to generate the most comprehensive summary statistics possible. The new variables and the methodology are outlined below and include the baseline and ending statistics produced in our school-year assessment.

Reading Level Variables

The Aspire data set included each student's reading levels in all four quarters of the school year in which they participated in the Aspire program (variables: *q1read, q2read, q3read,* and *q4read)*. Within the data set, students' reading scores were not coded universally, but rather by a two-digit integer, formatted as *G.X*, in which *G* represented the grade level the student was reading at, and the *X* following the decimal point was a number 1-3, representing whether the student was below average (1), sufficient (2), or excelling (3) at that grade level.

As some students in the sample were up to two grade levels behind in their reading assessment, the team introduced two new variables, *startread* and *endread,* to signify each student's performance in quarters one and four, respectively, including whether the student was one or two years behind in reading level, and whether the student was below average, sufficient, or excelling at his or her current grade level. To calculate each student's adjusted reading level, we recoded the student's reading scores for quarters one (*q1read)* and four *(q4read)* of their respective school years according to the following categories:

-3	*Two or more grade levels behind (G minus 2, regardless of X = 1 or 2)*
-2	*One grade level behind (G minus 1, regardless of X = 1 or 2)*
-1	*Slightly below average for grade level (G.1)*
0	*Sufficient for grade level (G.2)*
1	*Above average for grade level (G.3)*

Following this recoding, the team subtracted the adjusted first quarter score from the fourth quarter score to find the difference (*readchange),* which was then used to find the average change in reading attainment for Aspire students from quarters one to four of the students' respective school years.

Homework Score Variables

The Aspire data set also included each student's homework score for all four quarters of the school year that they participated in the Aspire program (variables: *q1hw, q2hw, q3hw,* and *q4 hw)*. Aspire logged the students' homework scores on the following scale:

60	*Unsatisfactory*
70	*Improving*
80	*Satisfactory*
90	*Outstanding*

Similar to how the team assessed reading level attainment, we created a new variable, *hwchange*, to indicate the difference between the student's homework scores for quarters four (*q4hw)* and one *(q1hw)*. We used this variable to find the average change in Aspire students' homework scores over the course of the school year.

Table 1 indicates the average baseline and concluding reading and homework scores of the students in the sample.

Table 1
Average Baseline and Concluding Reading Level and Homework Score by Grade Level

Quarter 1 & Quarter 4						
	N	Return. Students	Avg. Q1 Reading Level	Avg. Q4 Reading Level	Avg. Q1 Homework Score	Avg. Q4 Homework Score
First Grade	1	0	-2	0	80	80
Second Grade	2	0	-2	-1	75	85
Third Grade	16	0	-1.56	-0.5	80.6	81.2
Fourth Grade	27	12	-1.22	-0.67	77	75.6
Fifth Grade	36	28	-1.78	-0.92	78	78.6

New Research Question and Hypotheses

Although the Georgetown team cannot conduct a regression to determine a statistically significant estimate of the impact of students' attendance in the Aspire program on their reading scores, we can use descriptive data to indicate how students' reading levels and homework scores may have improved throughout their respective school years. The following research question and hypotheses will guide the discussion of descriptive statistics in the next chapter.

New Research Question: *How have students' reading levels and homework scores changed in the school year during which they participated in Aspire?*

Hypotheses

- **Hypothesis 1:** Students' reading scores will increase throughout the school year in which they participated in the Aspire program.
- **Hypothesis 2:** Students' reading gains will vary with the number of years they have participated in the Aspire program; moreover, students newer to the program will make more substantial reading progress.
- **Hypothesis 3:** Returning Aspire students will have, on average, more reading scores at or above their present grade level during the fourth quarter of that academic year than students new to Aspire.
- **Hypothesis 4:** Students' homework scores will increase throughout the school year in which they participated in the Aspire program.
- **Hypothesis 5:** Students' homework increases will vary with the number of years they have participated in the Aspire program; moreover, students newer to the program will make more substantial progress in regard to their homework scores.

IV. Data Analysis and Discussion

As noted, since the Georgetown team lacked the data required to conduct propensity score matching or a statistically-sound OLS regression, the team instead employed descriptive data analysis to assess the effects of the Aspire program on student participants.

Analysis

By generating basic statistical summaries of the data, including the newly created variables discussed in the methodology section (*startread, endread, readchange, q1hw, q2hw, q3hw, q4hw,* and *hwchange)*, the Georgetown team was able to assess the change in students' reading and homework scores from the start of the school year with the Aspire program, to the end of the year in the program. We generated overall statistical summaries for the 82 students in the sample, and summaries by subgroups (such as grade level or year in the Aspire program) and used these summaries to answer the research question and test the five hypotheses.

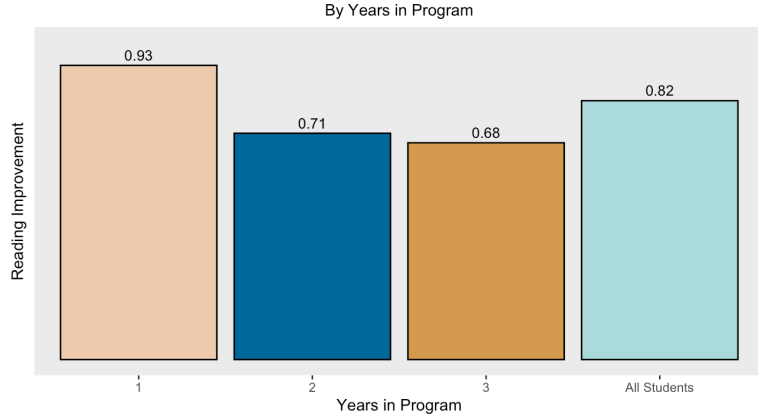
Each hypothesis is listed below, followed by the statistical finding and analysis based on our sample data:

Hypothesis 1: **Students' reading scores will increase throughout the school year in the Aspire program.**

Finding: The average Aspire student increased by 0.82 reading categories throughout his or her school year.

As noted, the Georgetown team coded reading scores to indicate whether a student's reading abilities were: behind by two grade levels (-3), behind by one grade level (-2), below average for their grade level (-1), sufficient for their grade level (0), or exceeding their grade level (1). As shown in Figure 1 on average, Aspire students' reading scores—for new, returning, and all students—improved nearly a full category during one year in the Aspire program.

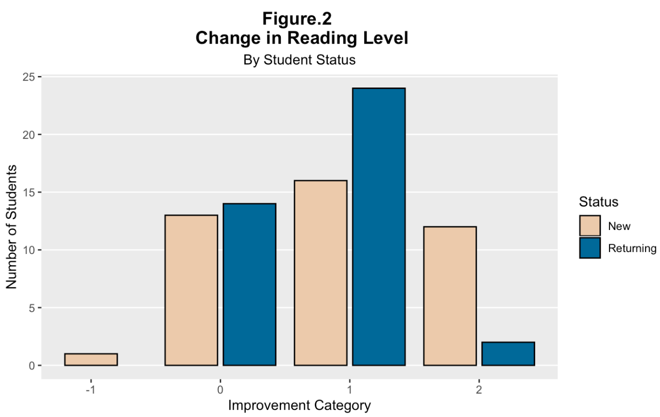
Figure.1
Change in Reading Level
By Years in Program

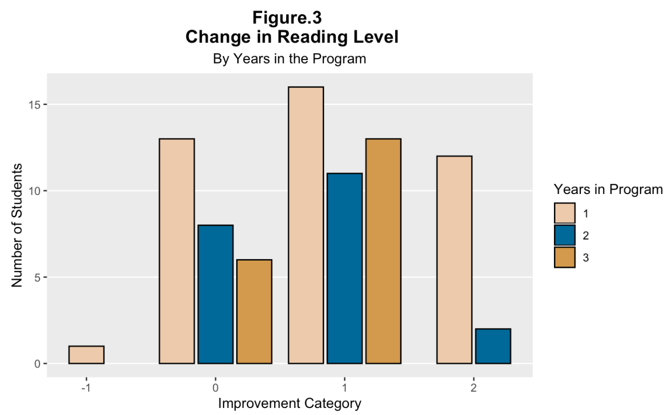


Hypothesis 2: **Students' reading gains will vary with the number of years they've participated in the Aspire program; moreover, students who are newer to the program will make more substantial reading progress.**

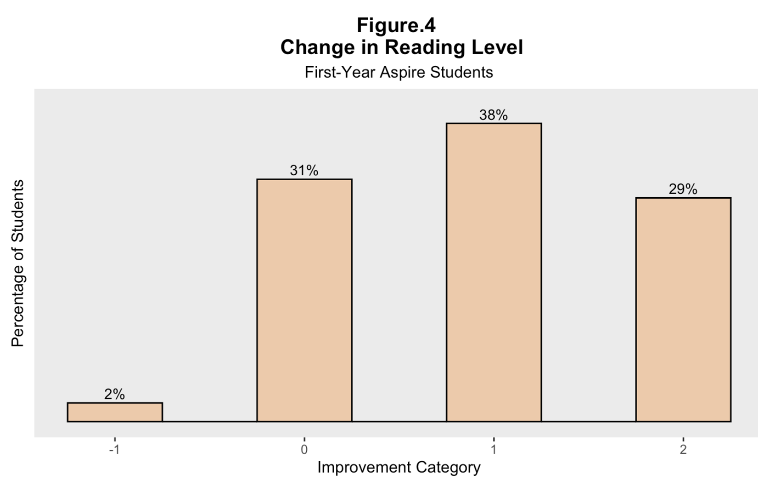
Finding: Student reading scores increased for both new and returning students.

Distribution of Aspire students: Figure 2 shows the change in reading level by student status ("New" = students in their first year of the program and "Returning" = students in their second or third year of the program). Figure 3 further provides these data by total number of years in the program.



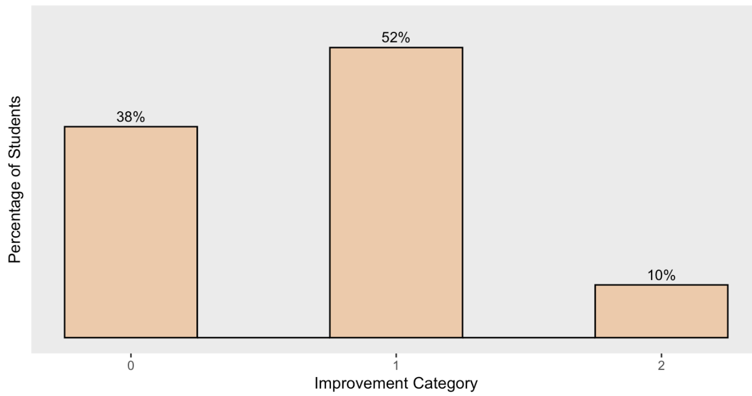


First-year Aspire students: Out of the 42 students in the sample who were new to the Aspire program, nearly 70 percent experienced a reading improvement of either one or two categories (38 percent experienced a one-category improvement and 29 percent experienced a two-category improvement). This is shown in Figure 4.



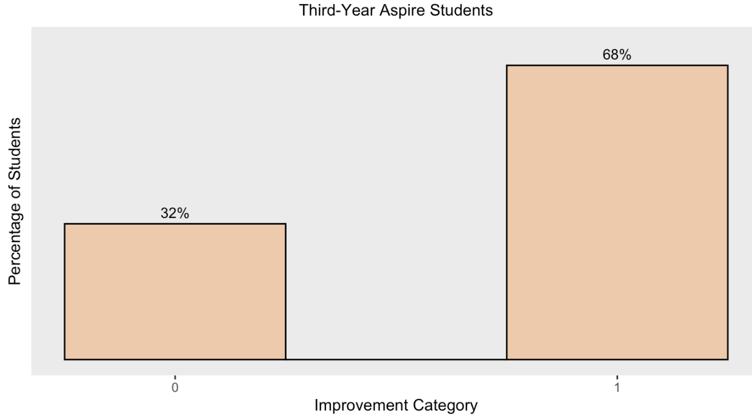
<u>*Second-year Aspire students:*</u> Figure 5 shows that out of the 21 students who had returned to the Aspire program for their second year, roughly 62 percent experienced a reading improvement of either one or two categories (52 percent experienced a one-category improvement and nearly 10 percent experienced a two-category improvement). Thus, more than half of second-year Aspire students continued to make substantial gains in their reading levels.

Figure.5
Change in Reading Level
Second-Year Aspire Students



<u>*Third-year Aspire students*</u>*:* Figure 6 shows that out of the 19 students who had returned to the Aspire program for their third year, 68 percent experienced a reading improvement of one category; the other 32 percent of their third-year peers showed no improvement. Overall, more than two-thirds of third-year Aspire students continued to experience improvement in their reading levels. This could indicate that a 'plateau theory,' in which students' progress declines throughout their time in the program, may not apply to Aspire students. However, more data would be needed to test this theory.

Figure.6
Change in Reading Level
Third-Year Aspire Students

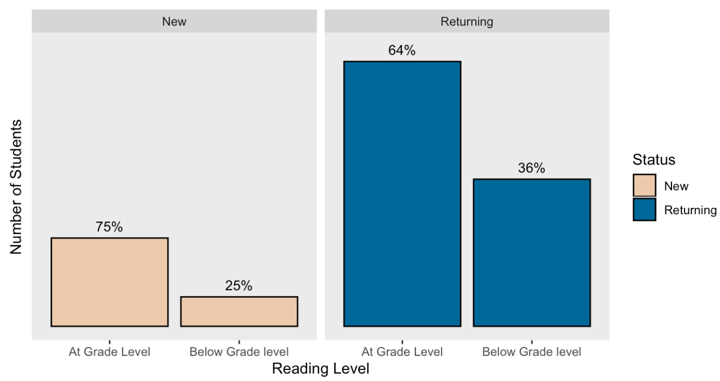


<u>Hypothesis 3:</u> **Returning Aspire students will have, on average, more reading scores at or above their present grade level during the fourth quarter of the academic year than students who are new to Aspire.**

Finding: The analysis fails to support or refute this hypothesis; more information is needed for an accurate assessment. The data analyzed for this hypothesis was restricted to fourth- and fifth-graders, as students from these grade levels were present as both new and returning students.

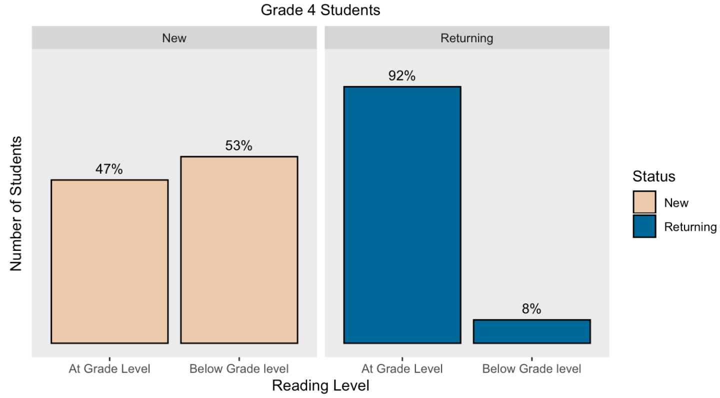
Figure 7 shows that, despite small and varied sub-sample sizes for each group, 75 percent of the first-year Aspire students in the fifth grade had roughly a fifth-grade reading level as opposed to 64 percent of returning fifth-grade Aspire students. However, this variation could possibly be attributed to the first-year students' low sub-sample size of 8 fifth graders versus 28 for returning fifth-grade, since more variation would be expected within a larger sub-sample.

Figure.7
Quarter 4 Reading Level
Grade 5 Students



In contrast, among fourth graders, roughly 92 percent of returning Aspire students were at or above a fourth-grade reading level, while only about 47 percent of new Aspire students could report such a level. This is depicted in Figure 8. These mixed results point to the need for a larger sample, in which the proportion of new students to returning students is approximately even.

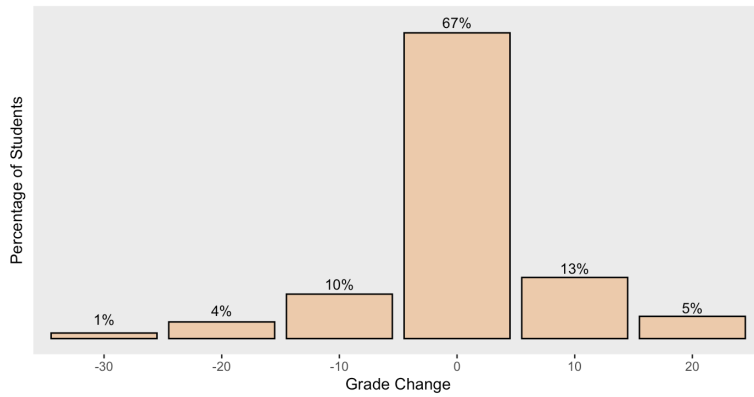
Figure.8
Quarter 4 Reading Level
Grade 4 Students



Hypothesis 4: **Students' homework scores will increase throughout the school year in which they participate in the Aspire program.**

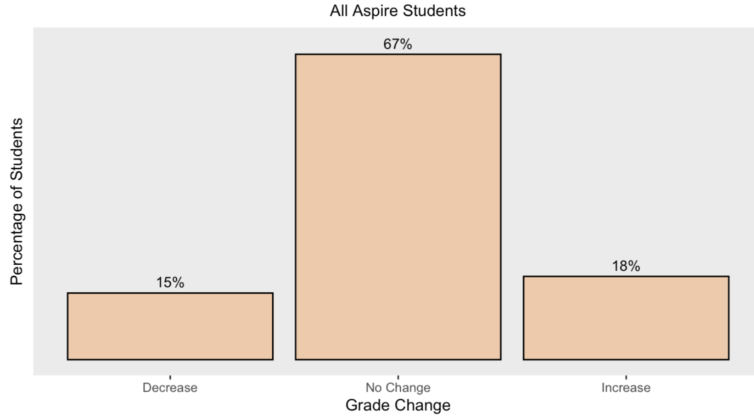
Finding: The analysis does not support this hypothesis. Although students' reading scores overwhelmingly increased, the majority (67 percent) of students' homework scores remained stagnant, as shown in Figure 9.

Figure.9
Homework Grade Change
All Aspire Students



However, of the remaining students, 18 percent saw increases in their homework scores, while 15 percent of students that saw reductions in their homework scores. This is shown in Figure 10.

Figure.10
Homework Grade Change
All Aspire Students



This information may be more reasonably interpreted by Aspire, as the Georgetown team recognizes that homework assignments may grow more challenging throughout the school year, in which case students maintaining or increasing their scores (a total of 85 percent across both categories) would certainly be an achievement. Regardless, homework assignments appear to be an area in which Aspire may consider bolstering its efforts as part of the program.

<u>*Hypothesis 5:*</u> Students' homework increases will vary with how many years they've participated in the Aspire program; moreover, students who are newer to the program will make more substantial progress in regard to their homework scores.

Finding: New students have a higher average fourth-quarter homework score and a higher percentage of improvements on homework scores throughout the school year; despite the distribution of homework scores for new and returning Aspire students taking a similar shape. This is shown in figures 11 and 12.

Figure.11
Homework Grade Change
By Student Status

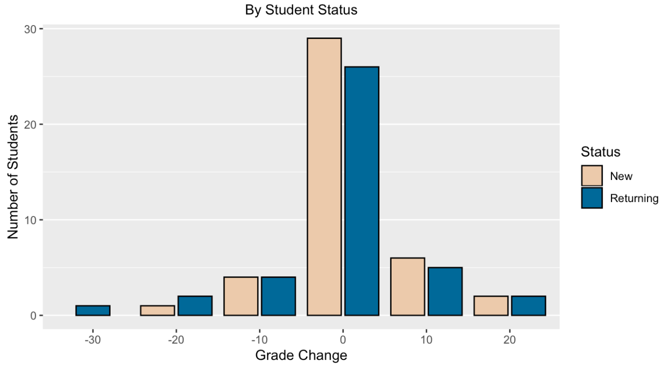
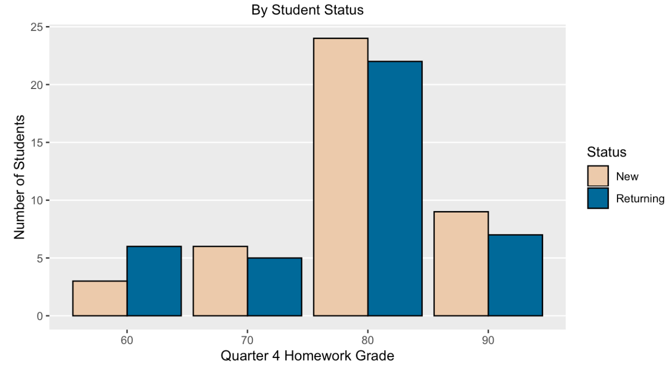


Figure.12
Quarter 4 Homework Grades
By Student Status



Specifically, new students showed an average fourth-quarter homework score of 79.3 and an average improvement of 0.95 points on homework scores from quarters one to four, as shown in figures 13 and 14. Alternatively, returning students showed an average fourth-quarter homework score of 77.5 and an average reduction of 0.5 points on homework scores between quarters one to four.

Figure.13
Average Homework Grade Change
By Student Status

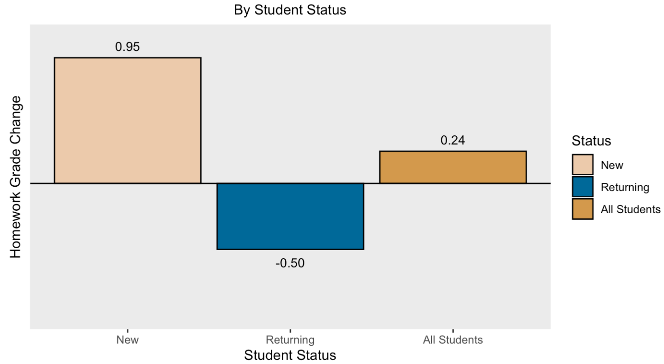
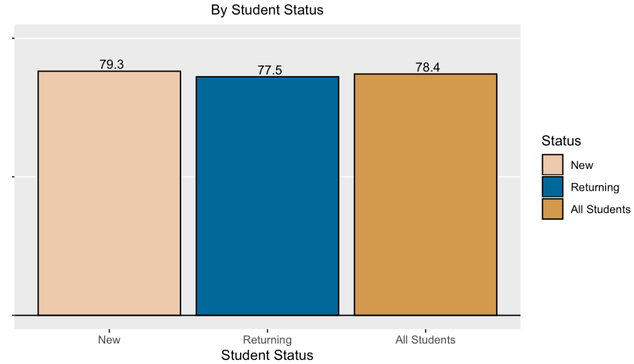


Figure.14
Average Quarter 4 Homework Grades
By Student Status



Summary

For the sample of 82 Aspire students, with complete data, observed from 2016-2019, the Georgetown team found the following based on descriptive analyses:

- On average, students' reading scores increased throughout the year while enrolled in the Aspire program;
- More than two-thirds of new Aspire students made substantial reading progress over the course of the year, however, returning Aspire students also showed widespread improvements;
- Mixed results exist as to whether new or returning Aspire students are more likely to have reading levels at or above their current grade levels;
- Although students' reading outcomes improved throughout the year, most students' homework scores remained unchanged between quarters one to four;
- New Aspire students made more significant progress with regard to improving their homework scores, and also had higher homework scores by the fourth quarter—whereas returning Aspire students, on average, saw lower homework scores and a slight decline in homework scores by the fourth quarter.

Further research is needed to validate the above finding; preferably by conducting a propensity score matching study—which could compare the outcomes of Aspire students to demographically-similar, non-Aspire students—or a statistically significant regression analysis. Only then may causation be implied between the Aspire program, and students' reading and homework outcomes.

V. Alternatives and Recommendations for Evaluating Aspire!

The research team was unable to conduct an impact evaluation due to insufficient data, but this chapter provides a framework that Aspire can use for future impact and implementation evaluations.

Recommendations for Conducting an Impact Analysis

To ascertain the impact of Aspire's after school program on its students, more data, and different types of data should be collected. This section provides suggestions on 1) data requirements to allow for a more robust data analyses, and 2) analysis recommendations, such as conducting quasi-experimental analyses, that can clearly identify Aspire's positive impacts on students' reading achievement as compared to similar students who do not receive this intervention.

Data requirements for conducting impact analyses

Aspire currently collects several key variables necessary for the conduct of an impact analyses. These include student-level data on:

- report card reading scores collected throughout the year
- days in attendance
- grade level
- school name
- English learner status
- special education status
- student of color status
- returning student or new student

At the school-level, Aspire asserts that almost all of the students they serve receive free or reduced-price lunch and live in affordable housing communities.

We recommend collecting additional variables and/or information which would allow for an impact evaluation, reduce selection bias, and allow for the construction of a more accurate regression model. These variables will serve as important control variables. Controlling for other potential impacts for student achievement helps ensure that results are attributed to the program itself rather than other factors.

We suggest that Aspire collect data on the following student-level control variables:

Gender: Researchers have found gender achievement gaps in math and English Language Arts subjects which vary by district according to the socioeconomic and gender makeup of a districts (Reardon et al, 2018). The average school district in the US has an English Language Arts gap that favors females. To control for this gap, Aspire may wish to collect and code gender of each Aspire student.

Student Race/Ethnicity: While we already know that Aspire serves low income students and students of color, the data we received does not explicitly identify student race or ethnicity. Different academic outcomes among different race/ethnicities is important to analyze.

Standardized reading level: Standardized test scores are the best way to compare Aspire students to their non-Aspire counterparts. In addition, comparison between pretest and posttest reading scores can effectively assess student progress throughout the year. The Georgetown team was unable to evaluate Aspire students' reading report card scores and compare them to their non-intervention peers because this score is dependent on several subjective factors, including: the reading material being used to teach the student, the teacher's evaluation of the student, and the assessment given by the teacher. The Standards of Learning (SOL) is an ideal standardized test. It includes a statewide annual assessment in reading in grade 3-8, and the outcomes have been well documented in the VDOE database. This allows for comparison between Aspire students and their peers at school, district, and state levels.

School Proportion of Free/Reduced Lunch Students: Aspire deliberately targets students with low socioeconomic status. However, we do not have information on these students' schools' socioeconomic status. This variable can be proxied by looking at the schools' proportion of students receiving free or reduced-price lunch. As defined by the National Center for Education Statistics (NCES), high poverty schools are schools where more than 75% of students receive Free or Reduced Price Lunch (FRPL); mid-high poverty schools are schools where 50.1%-75.0% of students receive FRPL; mid-low poverty schools are schools where 25.1%-50.0% of students receive FRPL; and low-poverty schools are schools where 25.0% or less of students receive FRPL (U.S. Department of Education, 2019).

According to Aspire's data set, 70% of students at Carlin Spring Elementary are eligible for free lunch and 10% of students are eligible for reduced price lunch; at Oakridge Elementary School 20% of students are eligible for free lunch and 5% are eligible for reduced price lunch (*Public School Review,* 2020). While most Aspire students are in mid-high to high poverty schools, Aspire also serves some students in low-poverty schools. However, any OLS regression analysis evaluating Aspire should control for Aspire's students' school's socioeconomic status because a low-poverty school is correlated with other factors, such as higher school resources.

School Racial Composition: In assessing students' academic outcomes, it is important to look at their school ethnic and racial composition. Schools often remain segregated by race and social class even after *Brown v. Board of Education*, the landmark Supreme Court decision that found racial school segregation to be unconstitutional. This de facto segregation is a majorly negative influence on students' academic outcomes. Having an integrated and diverse class full of peers helps create a stronger sense of academic self-efficacy (Schneider, 2017). A measurement of this variable would be a useful control in a regression mode evaluating Aspire's performance.

Parental Education: Parental educational level is a strong predictor of children's academic outcomes. Even when controlling for socioeconomic status and a child's intelligence, parental education has a strong association with positive, long-term academic success (Dubow et al., 2009). Therefore, it would be helpful to control for parental education levels in any model evaluating Aspire's performance.

Preschool Attendance: Preschool attendance helps to produce positive academic and socio emotional outcomes for students entering kindergarten. However, research has yet to establish a lasting impact of preschool attendance on academic outcomes through elementary school (Phillips et al., 2017). While Aspire works with 3rd to 5th graders, not kindergarteners, it may be useful for Aspire to collect data on long term effects of preschool attendance.

While a plethora of additional variables exist that can help predict student outcomes, we have focused solely on those described above to avoid multicollinearity, which can negatively impact statistical significance.

Principles of data collection

When collecting data, Aspire should consider the four CART principles—credible, actionable, responsible and transportable (Gugerty & Karlan, 2014):

- Credible: refers to collecting only data that accurately reflect the indicators that the program is trying to capture. For example, Aspire seeks to close the achievement gap between Black and Hispanic students and their White peers. However, the indicator "Student of color" is not accurate enough because it could potentially include Asian students making it difficult to conduct subgroup analysis of Hispanic and Black students separately. We recommend that Aspire specifies a student's "Race/ Ethnicity" more accurately, changing the definition to "Black/ Hispanic". This will allow for detailed analyses comparing achievement of Hispanic and Black students to the achievement of White students, and also allow for a comparison of Hispanic students to Black students.
- Actionable: refers to collecting only the data to be used in the evaluation. Aspire's 2018-2019 spreadsheet indicates that Aspire sought to collect students' reading scores quarterly. However, due to missing value, it is not feasible to use data for all four quarters in a regression model. We recommend that Aspire need only collect pre-test and post-test reading scores of its participants instead of four quarters.
- Responsible: refers to an awareness of falling into the trap of being "data-hungry." Overreaching/overcollecting can compromise the quality of the data as well as the ability to analyze it. We recommend a thoughtful approach to data collection, making certain that data collected on every variable has a purpose towards measuring a component of the program or serving as a control variable.
- Transportable: refers to transportability of existing findings to other programs and contexts. In the context of Aspire, it also means to objectify the data so that it can be compared to data on other students at school, county and even state level. For example, there is a mismatch between how reading score data are collected/reported by APS, VDOE, and Aspire. Aspire collects students' reading scores from school report cards (scale from grade level ± 0.1~0.9). The APS and VDOE database report students' reading levels (include fail, pass, proficiency, advanced). Without a transportable standardized test score, it is difficult to make accurate comparisons between Aspire participants and non-participants.

Successful quantitative analysis and strong evidence rely heavily on data quality. Table 2 provides recommendations based on the literature for ensuring high-quality data.

Table 2
Indicators of high-quality data

Quality Indicators	Definition
Validity	The accuracy of data measurement. Data represent the variables they are intended to capture. A measurement is considered "valid" if it is both accurate and precise.
Reliability	The consistency of the data measurement, and whether the research method produces stable and consistent results.
Completeness	The degree to which all required data are available in the dataset.
Precision	Sufficient details are included in the data.
Integrity	Data are not manipulated or affected by deliberate bias or subjective factors.
Timeliness	Data and information are timely and regularly updated.

Source: Peersman, 2014.

Recommendations for Conducting an Implementation Evaluation

Roadmap for conducting a program evaluation

Evaluation is a crucial component in any program intervention. Generally, the evaluation seeks to answer three types of questions (Imas & Rist, 2009):

- Descriptive questions: descriptions of what is taking place, including study context, organizational relationships, implementation process, stakeholder views, etc.
- Normative questions: comparisons between what has been done and what should be done. It can also be used to examine the relationship between inputs and outputs, and whether the targets of the program are accomplished or not.
- Cause-and-effect questions: examination of the program outcomes. Typically, it assesses the effectiveness of the program intervention.

Researchers typically use two evaluation approaches to program evaluation: implementation evaluation and impact evaluation. Table 3 reveals the differences between implementation evaluation and impact evaluation.

Table 3

Differences between implementation evaluation and impact evaluation

	Implementation evaluation	Impact evaluation
Answer question	Descriptive question	Cause-and-effect question
Focus	Program implementation and improvement	Program outcomes
Period	During the implementation phase of a project	At the end of an intervention
Research method	Qualitative: descriptive analysis	Quantitative: randomized experiment or quasi-experimental analysis
Evaluators	Internal evaluation: conducted by people within the organization	External evaluation: conducted by people from outside the organization (mostly)

Both types of evaluation are necessary for an organization to evaluate its program at different stages. The evaluation framework of the Institute of Education Sciences (IES) has identified four typical stages of a project, which was presented in Table 4 (IES, 2012). This section provides further information about conducting an implementation evaluation which may be useful to Aspire.

Table 4
IES Evaluation Framework: Four typical stages of program evaluation

IES evaluation category	Stage of program development	Research questions	Audience/ evaluator	Evaluation design(s)/ approaches	Evaluation products
Stage 1 Exploratory	Idea generation	Is my idea likely to work? What has already been tried? Is there a demand for my idea?	Program developer is the audience/ evaluator is often internal	*Descriptive*: Needs assessment	*Informal and iterative*: Literature review, description of the need
Stage 2 Development and innovation	Program being developed and implemented in limited number of sites	How is the program working? Have I got the right components? How can it be improved?	Program developer is the audience and possibly also funding agencies/ evaluator can be internal or external	Mostly *descriptive*: Developmental evaluation, design-based research, pilot studies, evaluability assessment	*Informal and iterative*: logic model, program description, implementation feedback, initial evidence of promise, evaluability findings
Stage 3 Efficacy and replication	Program is fully developed, not yet tested	Is the program effective under ideal conditions? Does it need further refinement? Is it ready for wider dissemination and testing?	Audience is broader. Often program developer, funding agencies: Federal, state and local decision-makers/ evaluator is external	*Experimental/ quasi-experimental:* including implementation data feedback to program developers to improve performance	*Formal*: *some iterative*: Report on program impacts including identification of key active ingredients, ongoing report of implementation data
Stage 4 Scale-up	Program is fully developed with some indication of the effectiveness	Is the program effective under real-world conditions?	Audience is broader. Often program developer; funding agencies; federal, state and local decision-makers/ evaluator is external	*Experimental/ quasi-experimental:* Implementation data used to help understand findings (is not feedback to program developers)	*Formal*: Report on program impacts including identifying for whom these program works, and under what conditions, and implementation information.

Aspire currently has four learning sites and approximately 100 participants. It has offered various reading instructions to develop students' reading skills, and is now developing other classes such as mathematic and socio-emotional activities. We assume that Aspire is at Stage 2 of the IES evaluation framework (Development and innovation), transitioning towards Stage 3 (Efficacy and replication).

Therefore, before reaching the stage of impact evaluation, Aspire can do more descriptive research to gain a comprehensive view of how its program works.

Implementation evaluation

Conducting an impact evaluation can methodologically determine the effectiveness of the Aspire intervention, but it requires a robust sample size and a comparison group. However, much can be learned by conducting an implementation evaluation of a program either in leu of, or in conjunction with an impact evaluation. We recommend that Aspire considers conducting an implementation evaluation. This will require collection of data on additional variables, which can be expanded based on what aspects of the program implementation Aspire wishes to assess.

Implementation Evaluation

Implementation evaluation can effectively monitor staff performance and gather details of operations. The indicators collected during the implementation help identify the advantages and disadvantages of a program, which enables the program developers to revise and improve the intervention in the future.

An implementation evaluation is driven by a program's theory-of-change model— the inputs, activities, and outputs of the program. The Aspire program consists of 15 hours of instruction each week and a six-week intensive summer camp, including academic and socio-emotional classes. Aspire employs an innovative, multilateral approach to program implementation and utilizes the returning middle school volunteers as "peer mentors" to its younger students. Each of these aspects of the program can be measured separately or together to evaluate how the program is implemented and look for ways to improve it. We recommend the following variables:

Program attendance: refers to daily program attendance records of the participants. Program attendance is an important indicator of the dosage of the intervention. Aspire collects this variable already.

Program content: refers to the activities included in the afterschool program, the amount of time allocated to each activity, and the curriculum structure (Mahoney et al., 2007). Aspire collects whether a student has received both academic (reading and math) and socio-emotional support, but we do not know about the content of this impact. Details of this content as well as program operations can help stakeholders and evaluators understand more clearly about what takes place in the Aspire program.

Program engagement: refers to whether students are actively engaged in the program. A common observation tool for evaluating practices in after-school programs is the Promising Practices Rating System (Vandell et al., 2004). This system includes different perspectives, such as *engagement, supportive relations with adults* (the program instructors), and *supportive* *relations with peers*. (For details about the Promising Practices Rating System, see Appendix III: Evaluation Tools.)

Program feedback: Effective partnerships between schools, families, and peers provide valuable assets to an afterschool program. Considering the unique advantage of Aspire in closely collaborating with schools, parents, and students, collecting feedback and suggestions from these stakeholders may be of interest to Aspire, as a mechanism for future program improvement.

Geiger and Britsch (2018) have provided comprehensive focus group questions and surveys (including student participants, teachers, staff and parents) as tools for program evaluation. We have suggested several evaluation tools which Aspire may find useful (see Appendix III: Evaluation Tools).

Recommendations for Measuring Socio-Emotional Learning (SEL)

Our literature review indicates that children living in housing and neighborhoods with poor physical quality have lower socio-emotional health than other children, and that SEL is associated with increased academic achievement. Research indicates that SEL helps students transition to secondary school, decreases aggressive conduct, and increases helping behavior. SEL has also been shown to be effective in other contexts, including afterschool programs. The literature shows that high levels of socio-emotional wellbeing appear to be associated with a strong connection to mentors. Our discussion of the SEL literature in the literature review provides Aspire with the building blocks necessary to monitor SEL in the future, highlighting ways that Aspire can gather SEL information from their students.

SEL activities can take many forms; there is no one agreed method of conducting SEL and the research field continues to develop. Activities that contribute to SEL can range from prosocial sessions, where children actively engage with one another, to exercises aimed at developing emotional regulation and perspective-taking skills. It is important, therefore, to outline the different forms that SEL programs can take and how Aspire can leverage its resources. Aspire already engages in some SEL, such as prompting students to express their feelings with more vocabulary and associating feelings with different colors. To assess the impact of its current SEL efforts on students, Aspire should consider: 1) what additional SEL activities it could implement with students, 2) standardizing all SEL-related activities so that every instructor is conducting them in the same way and with the same amount of dosage, and 3) systematically measuring students socioemotional behaviors to determine the short-term impacts of its SEL efforts. We describe these steps in more detail in the following paragraphs.

Step 1: Deliver On-Site SEL Sessions

In an educational setting, SEL programs can be delivered as a separate class, where students can engage in a specifically curated curriculum that is an add-on to their normal academic curriculum. This method has been used in various schools with students required to attend. SEL programs can also be integrated with regular classes, where socio-emotional concepts are incorporated during regular class times as a supplement. In the case of former, the separate, dedicated SEL classes, the material can be delivered by existing teachers or by external staff. Research indicates that employing external staff can add considerable financial costs and therefore training existing staff to implement the SEL may be the most cost-effective route.

Aspire has the advantage of being involved in their students' journey in different environments; at school, home, and afterschool at Aspire locations. This gives Aspire multiple options on how and where to best engage students. Given that students spend a considerable portion of time at Aspire locations and Aspire has control over activities that students engage in at these locations, we recommend delivering SEL sessions on-site, using existing staff.

Step 2: Standardize SEL Activities

To better measure the impact of SEL activities on students' socio-emotional behaviors, it will be important for Aspire to standardize all SEL activities to the extent possible, including training the staff on how to implement the activities in the same way. It will also be important for Aspire to determine and administer the proper dosage of SEL for impact. For example, should students receive an SEL activity once a day, weekly, or monthly? It may depend on the activity, but the dosage of instruction may have varying impacts. So, dosage will be an important variable to standardize and measure.

Offering SEL as Distance Learning

Due to the current situation related to coronavirus, holding on-site SEL sessions is not possible. Some organizations have collated best practices on holding *online* SEL sessions. The Institute of Social and Emotional Learning posts blogs that include specific examples of SEL distance learning, including check-ups, activities students can do while following instructors, and activities they can do by themselves (Institute for Social and Emotional Learning, 2020). A link to their website and blog posts is in the bibliography for reference.

Step 3: Collect and Evaluate SEL data

We recommend that Aspire starts collecting meaningful data relating to socio-emotional skills before engaging in systemized SEL activities. This will require, first, creating a baseline assessment for each incoming and new student. Next, Aspire should collect follow-up data from students once or twice per year, perhaps at the end of the fall semester and then at the end of the spring semester. Collecting follow-up data will help Aspire adjust the implementation of SEL activities and also provide data on student growth related to socioemotional behaviors.

One way to structure SEL activities and their respective monitoring is through assessing five categories (Durlak et al. 2011). Table 4 provides a detailed description and examples of evaluation in these categories, as well as purchasing options for useful monitoring methods.

- Social and emotional skills
- Attitude towards self and others
- Positive social behavior
- Conduct problems
- Emotional distress

Social and emotional skills: This includes an assessment of emotions from social cues, interactive problem solving, decision making and conflict resolution. Monitoring this category can rely on data of self-reports from students, teachers, or parents, which can be collected through questionnaires, role plays, or interviews. Teachers can rate students in everyday situations, such as their ability to work with others or control their anger.

Attitudes toward self and others: This includes how positive the student views themselves (self-esteem and self-efficacy), their school (behavior towards teachers and the school), and various social topics (violence, drug use). Similarly, data can be collected regularly by Aspire through self-reporting by students in the form of a short and simple questionnaire or assessment.

Positive social behavior: This refers to daily behavior such as playing and talking with others. To measure this category, researchers often use Elliott and Gresham's Social Skills Rating Scale (Elliott et al. 1988), where teachers can systematically assess the social behaviors of students.

Conduct problems: This includes behavioral issues such as bullying, school suspensions, disorderly class behavior, and general noncompliance. Having school records data on school suspensions and expulsions will allow for measurement of this category, along with other measures used by researchers such as the Child Behavior Checklist (CBC), Youth Self-Report (YSR), or the Teacher's Report Form (TRF).

Emotional distress: This category includes measuring the mental health of students, including depression, anxiety, and stress. This can be self-reported by students, parents, or teachers using the Children's Manifest Anxiety Scale (Kitano, 1960).

Aspire is in a strong position to actively measure any or all of these socioemotional categories. The active involvement of Aspire staff with students directly and frequently enables it to create a harmonic monitoring system that can greatly benefit their students' socioemotional well-being and impact their academics. Aspire should begin collecting SEL data for the purpose of impact evaluation and student monitoring, and a systemized measure will ensure greater validity of future program evaluations.

Table 5
Monitoring SEL activities

Category	**Definition**	**Monitoring Process**	**Notes**
Social and emotional skills	Assessment of emotions from social cues, interactive problem solving, decision making and conflict resolution	Self-report: student/ teacher/ parent, through questionnaires, role plays or in-depth interviews; Observational study: teachers can rate students in everyday situations	
Attitudes toward self and others	How positive the student views themselves (self-esteem and self-efficacy), their school (behavior towards teachers and the school) and various social topics (violence, drug use)	Self-report: students	
Positive social behavior	Daily behavior such as playing and talking with others, and not answers to hypothetical questions	Elliott and Gresham's Social Skills Rating Scale	
Conduct problems	Behavioral issues such as bullying, school suspensions,	School suspension data Partnership with the Child Behavior Checklist	The checklist, along with other monitoring methods, can be

	disorderly class behavior and general noncompliance	(CBC), Youth Self-Report (YSR) or the Teacher's Report Form (TRF)	purchased from ASEBA through their website https://aseba.org/
Emotional distress	Monitoring of mental health issues such as depression, anxiety or stress	Self-report: student/ parent/teacher Children's Manifest Anxiety Scale	Can be purchased from WPS through their website https://www.wpspublish.com/rcmas-2-revised-childrens-manifest-anxiety-scale-second-edition

Recommendations for Conducting a Longitudinal Study

Aspire has expressed interest in conducting a longitudinal study that assesses its students' long-term progress, including progress made after students conclude their Aspire experience. This section outlines steps for developing and implementing a longitudinal study.

Step 1: Establish District Data-Use Agreements

Districts often require research applications and signed agreements before providing student record data. Aspire should work with local school districts where students matriculate to high school to obtain data-use agreements.

Step 2: Collect and record data about Aspire students

Aspire may consider obtaining student records from their students' high schools and/or conducting a telephone or web-based survey to follow up with their students each year.

A web survey would directly collect data from students about their academic and professional intentions and achievements. The survey content would vary each year, but could include questions such as:

- Intent to graduate from high school
- Intent to enroll in a 4-year college/university or a vocational or 2-year degree program
- Self-reported GPA
- Self-reported in school and out of school activities (e.g., clubs and sports)
- Employment status
- College enrollment status
- Self-rating of personal motivation or goal-setting and other socio-emotional behaviors

Below is a list of variables Aspire could consider as a guide to data collection, to be obtained either from student records and/or a student survey. Each variable is given a variable name, a description, date of recommended collection, and is accompanied by suggestions for analysis. We recommend following up with students at the following benchmarks, which would follow typical graduation rates: five years (to assess junior high or high school performance), eight years (post-high school), and thirteen years (post-college and employment).

Variable	Description	When to Collect	How to Collect	Suggestions for Analysis
HSGPA	A categorical variable (coded 1.0 - 4.0) that indicates the student's high school GPA.	5-year mark 8-year mark	Student record, if possible	Generate basic summary statistics (mean, median, mode, standard deviation, etc.) as well as the full distribution of high school GPAs, focusing on the average GPA of former Aspire students.
HSEXCURR	A dummy variable indicating whether the student was involved in extracurricular activities in high school.	5-year mark 8-year mark	Survey	Generate the percentages of 0 and 1 to see whether or not the average Aspire student participated in extracurricular activities.
HSGRAD	A dummy variable (coded 0 for no and 1 for yes) that indicates whether the student graduated from high school.	8-year mark	Student record, if possible	Generate the percentages of 0 and 1 to see whether or not the average Aspire student graduated high school.
COLLAPP	A dummy variable indicating whether the student applied to a four-year college or university.	8-year mark	Survey	Generate the percentages of 0 and 1 to see whether or not the average Aspire student applied to a four-year college or university.
COLLACC	A dummy variable indicating whether the student was accepted to a four-year college or university.	8-year mark	Survey	Generate the mean/average to see whether or not the average Aspire student was accepted to a four-year college or university. Further, generate the average

				of this acceptance for those who applied (COLLAPP = 1) to know the percentage of acceptance among applicants.
COMCOLL	A dummy variable indicating whether the student enrolled in community college, for any length of time.	8-year mark	Survey	Generate the mean/average to see whether or not the average Aspire student graduated high school.
COLLGPA	A categorical variable that indicates the student's GPA at a four-year college or university.	8-year mark	Survey	Generate basic summary statistics (mean, median, mode, standard deviation, etc.) as well as the full distribution of college GPAs, focusing on the average GPA of former Aspire students.
COMCOLLGPA	A categorical variable that indicates the student's GPA at a community college.	8-year mark	Survey	Generate basic summary statistics (mean, median, mode, standard deviation, etc.) as well as the full distribution of community college GPAs, focusing on the average GPA of former Aspire students.
ASSOCDEG	A dummy variable indicating whether or not the student received his or her associate degree.	13-year mark	Survey	Generate the percentages of 0 and 1 to see whether or not the average Aspire student obtained an associate degree.
BACHDEG	A dummy variable indicating whether the	13-year mark	Survey	Generate the percentages of 0 and

	student received his or her bachelor's degree.			1 to see whether or not the average Aspire student obtained a bachelor's degree.
MASTDEG	A dummy variable indicating whether the student received his or her master's degree.	13-year mark	Survey	Generate the percentages of 0 and 1 to see whether or not the average Aspire student obtained a master's degree.
PHD	A dummy variable indicating whether the student received his or her doctoral degree.	13-year mark	Survey	Generate the percentages of 0 and 1 to see whether or not the average Aspire student obtained a doctoral degree.
PTEMPLOY	A dummy variable indicating whether the student is currently employed in a part-time capacity.			Generate the percentages of 0 and 1 to see whether or not the average Aspire student is employed in a part-time capacity.
FTEMPLOY	A dummy variable indicating whether the student is currently employed in a full-time capacity.	13-year mark	Survey	Generate the percentages of 0 and 1 to see whether or not the average Aspire student is employed in a full-time capacity.
ANNUALINC	A categorical variable indicating the student's annual income bracket (coded as 1 = $0-10,000; 2 = $11,000-20,000, etc.).	13-year mark	Survey	Generate basic summary statistics (mean, median, mode, standard deviation, etc.) as well as the full distribution of incomes, focusing on the average income of Aspire students, and the highest-obtained income.

Step 3: Conduct a Longitudinal Data Analysis

A longitudinal study can provide Aspire with a comprehensive view of each student's long-term academic and professional progress following the Aspire program. The main challenge with conducting a longitudinal study is sample attrition, or the rate at which students would "drop out" of the sample group by not responding and/or the rate at which their records might be lost. Also, if Aspire is already starting with a sample of less than 100 students, by the end of the study, the remaining respondent group could be too small to yield meaningful data on college performance or employment. However, if Aspire could not generate statistically valid conclusions using the collected data, it would at least be armed with summary statistics on the program's ability to yield positive long-term outcomes.

VI. Conclusion

The Georgetown Team found that Aspire students made substantial reading progress over the course of a school year, though students' homework scores remained unchanged. Further research is needed to validate the results we found via descriptive statistical analysis. Our analysis fails to demonstrate a strong implied relationship between the Aspire program and its students' academic outcomes. A propensity score matching method would best estimate the effects of the program by creating a counterfactual group based on shared characteristics. To apply such a method, future evaluation research should follow the data collection recommendations outlined in Section V.

To support a more comprehensive assessment of the Aspire program, the Georgetown Team also provided recommendations on adopting an implementation evaluation, developing social emotional learning activities, and establishing a longitudinal assessment of Aspire's academic impacts.

Minority and underprivileged students face many hurdles in obtaining a quality education, even when residing in an affluent school district. Aspire is committed to closing the achievement gap between underprivileged students who are disproportionally students of color and their more affluent peers. The Georgetown Team is grateful to have worked with Aspire in furthering their mission. We hope that this report will provide Aspire with a clear understanding of future steps to take that will allow them to improve academic outcomes for Arlington students.

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Appendix

I. *Abbreviations*

APS—Arlington Public Schools
ATE—Average Treatment Effect
BCA—Benefit-Cost Analysis
CBQ—Children's Behavior Questionnaire
EDS—Economically Disadvantaged Student
ESOL—English to Speakers of Other Languages
FRPL—Free or Reduced Price Lunch
HILT—High-Intensity Language Training
HLE—Home Literacy Environment
IEP—Individualized Education Program
NCES—National Center for Education Statistics
NWREL—Northwest Regional Educational Laboratory
NELP—National Early Literacy Panel
PPRS—Promising Practice Rating System
PSM—Propensity Score Matching
SEL—Socio-emotional Learning
SES—Socioeconomic Status
SOL—Standards of Learning
RCT—Randomized Controlled Trial
TANF— Temporary assistance for needy families
VDOE—Virginia Department of Education

II. *Table of Variables*

Dependent variables	Type	Level	Source	Time Frame
Reading Score	Continuous	Individual	Aspire	Year
Independent Variable				
Aspire program participation	Dummy	Individual	Aspire	Year
Aspire days attendance	Continuous (#days)	Individual	Aspire	Year
Control variables				
Race (Student of color)	Dummy (White; Non-White)	Individual	Aspire VDOE	Fixed
Economically disadvantaged (Proxy for low SES)	Dummy (Yes, No)	Individual	Aspire VDOE	Year
SOL score (English reading & English writing)	Continuous	Individual	VDOE	Year
Grade (Proxy for age)	Discrete (3rd to 5th grade)	Individual	Aspire VDOE	Year
English Language Learner Status (ESOL or HILT)	Dummy (Yes, No)	Individual	Aspire VDOE	Year
Special Education Status (IEP or 504)	Dummy (Yes, No)	Individual	Aspire VDOE	Year
School Socioeconomic Status	Continuous (percentage)	School	APS	Year
School Racial Composition	Continuous (percentage)	School	APS	Year
Suggested additional control variables				
Ethnicity	Categorical	Individual		
Family Structure (single or divorced parent)	Dummy (Traditional, Non-traditional)	Individual		
Parental Education	Categorical	Individual		
Preschool Education	Dummy (Yes, No)	Individual		
Gender	Dummy (Male, Female)	Individual		

III. *Evaluation tools*

Promising Practice Rating System (PPRS)

Brief description: PPRS is an observation tool designed for the afterschool program staff and administrators to rate promising program practices. It was first used in the Study of Promising After-School Programs, which was a national study of high-quality program targeting at disadvantaged children and youth. The overall PPRS forms include detailed information of:

- Activity context coding
- Activity description
- Activity promising practices rating form
- Program promising practices ratings & overall program quality

Link to source:
https://youthrex.com/wp-content/uploads/2019/10/Promising-Practices-Rating-System.pdf

Out-of-School Time Program Evaluation: Tools for Action

Brief description: Northwest Regional Educational Laboratory (NWREL) aims to provide research assistance to improve the results of high-quality educational programs. It has extensive experiences in developing surveys, working with school districts and evaluating the connection among programs, schools and community. This series of out-of-school program evaluation tools include:

- Student Participant Survey
- Spanish Student Participant Survey
- Student Participant Focus Group Questions
- Teacher Survey
- Parent Survey
- Spanish Parent Survey
- Staff Survey
- Staff Focus Group Questions
- Partnership Survey

Link to source:
https://educationnorthwest.org/sites/default/files/resources/ost_tools.pdf